SUHRID BEHARI PAUL

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Summary

AI researcher and engineer (B.Tech IT, CGPA 9.57/10) with research experience in agentic AI, LLM-based query optimization, and the robustness and security of data-driven systems in 5G/6G networks. Junior Research Fellow at IIT Guwahati. Building evaluation-first LLM systems that verify their own outputs, and now turning that toward empirical study of how tool-using AI agents fail and how safeguards help.

Experience

Junior Research Fellow July 2026 – Present
Indian Institute of Technology Guwahati, Dept. of Computer Science and Engineering

- Research predictive inter-RAT handover in 5G/6G networks and the security implications of data-driven prediction.
- Evaluate handover decisions on radio measurements using simulation-driven experiments.
- Investigate robustness of prediction mechanisms and potential vulnerabilities in next-generation wireless networks.

Research Intern, Query Planning with Agentic AI June–July 2025
New Jersey Institute of Technology, USA *Certificate*

- Developed a probabilistic query-optimization framework for personalized top-k retrieval using LLM-based approaches.
- Explored ranking strategies to reduce computational overhead in interactive query-planning systems.
- Presented the work internationally at the NJIT URI Research Symposium 2025.

Research Intern, Early Detection of DoS Attacks in 5G June–July 2024
Indian Institute of Technology Guwahati

- Explored data-driven approaches for detecting Denial-of-Service attacks in 5G networks.
- Performed network-security analysis to support attack detection and network resilience research.

Projects

Failure Modes and Safeguards in Tool-Using AI Agents Python, LLM APIs — GitHub
Independent Research Project — In Development

- Built a sandboxed harness (simulated tools; 5 tasks, 3 injection styles) scoring unsafe actions deterministically from tool-call logs, comparing a baseline agent against prompt hardening, input filtering, and layered safeguards.
- Reports unsafe-action rate with confidence intervals alongside task completion and false-blocking rate; evaluation on real models is the next step.

Agentic Research Analyst with Evidence Verification Python, Anthropic API, BM25 — GitHub
Independent Project — In Development

- Built a multi-stage agent (planner, retrieval, extractor, verifier, synthesizer, critic) that checks each claim against its cited passage, drops unsupported claims, and abstains when evidence is missing.
- Implemented a RAG baseline and evaluation harness (Recall@k, unsupported-claim rate, abstention accuracy, latency, cost) with tests.

Natural-Language-to-SQL Agent with Query Optimization Python, PostgreSQL, SQLGlot — GitHub
Independent Project — In Development

- Built a PostgreSQL agent with SQL validation, read-only execution, error repair, and clarification; the validator and the read-only role each blocked 11/11 destructive test statements.
- Benchmarked query rewrites with `EXPLAIN ANALYZE`: the planner was already near-optimal on most queries (rewrite helped on 2 of 12).

Technical Skills

Languages	Python, SQL, Java, C
AI & LLMs	Agentic workflows, LLM evaluation, adversarial robustness (5G context), retrieval, probabilistic query optimization
Data & Databases	Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Power BI
Tools	Git, GitHub, Streamlit, MATLAB

Education

Heritage Institute of Technology, Kolkata 2022 – 2026
B.Tech in Information Technology — CGPA: 9.57/10

Achievements & Leadership

- International Presenter, NJIT URI Research Symposium 2025.
- Hult Prize Campus Finalist (Top 5), Second Stitch initiative; Team Lead, Smart India Hackathon 2024 and Diversion Hackathon 2025.